ANIME RATING PREDICTOR

Machine Learning Project Report | Project ID: P45

Student Name	Dev Sharma
Roll Number	H-40
Enrollment Number	24004501210416
Division	E
Project ID	P45

1. Problem Statement

Anime platforms contain many titles with different popularity levels and community ratings. The aim of this project is to use anime metadata to classify whether a title is a “Top Anime”. The target is defined as top_anime = 1 when rating ≥ 8, otherwise 0. The task is binary classification, not prediction of an exact rating.

The model should learn from available metadata (episodes, members, genre and anime type), then estimate the class for unseen anime. A key challenge is that only a small fraction of records meet the Top Anime threshold, so evaluation must consider minority-class metrics such as recall and F1 in addition to accuracy.

2. Dataset Description

The project uses **anime.csv**, containing 12,294 rows and 7 columns. The separately supplied rating.csv was not used because the assignment instructions specifically say to use anime.csv only. The original Kaggle URL was not present in the supplied materials, so it is not reproduced here.

Data dictionary

Column	Meaning	Type / decision
anime_id	Unique anime identifier	Numeric; drop from inputs
name	Anime title	Text; drop from inputs
genre	Comma-separated genre labels	Text; encode top 15 genres
type	TV, Movie, OVA, etc.	Text; one-hot encode
episodes	Episode count	Numeric; retain after cleaning
rating	Community rating	Numeric; target creation only
members	Community member count	Numeric; retain

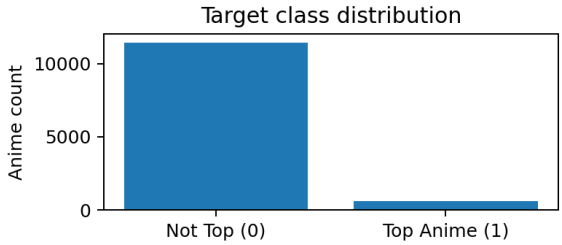
3. Data Cleaning and Exploratory Data Analysis

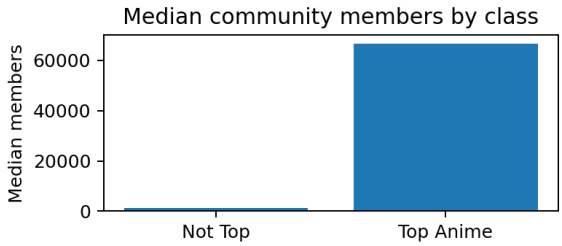
Rows with missing ratings were removed because a target class could not be assigned. The “Unknown” episode values were converted to missing and replaced with the median. Missing genre/type entries were filled with “Unknown”; anime_id and name were excluded from model inputs. The rating column was excluded from X to prevent target leakage.

Cleaning check	Observed result
Original shape	12,294 × 7
Missing rating rows removed	230
Rows after rating filter	12,064
Unknown episodes	188; replaced with median = 2
Missing genre after filtering	47; filled as Unknown
Duplicate rows	0
Missing values after cleaning	0

Best EDA charts and insights

The class distribution is strongly imbalanced: 11,437 Not Top Anime (94.8%) and 627 Top Anime (5.2%). This means accuracy alone can look high even when many Top Anime titles are missed.





Key insights: Top Anime has median members of 66,756, compared with 1,346 for Not Top Anime. Median episodes are 12 for Top Anime and 2 for Not Top Anime. Average genre_count is 4.05 for Top Anime versus 2.89 for Not Top Anime. Members has a positive association with the target (correlation ≈ 0.420); genre_count is weaker (≈ 0.154), while episodes is very weak (≈ 0.025). These are associations, not proof of cause.

4. Feature Engineering, Model and Settings Tried

A new feature, **genre_count**, counts the number of genres assigned to each anime. The 15 most frequent genres were converted to binary indicator features, and anime type was one-hot encoded. This produced 24 candidate features. SelectKBest with the ANOVA F-test selected 15 features using the training data.

Selected features: members, genre_count, genre_Action, genre_Drama, genre_Shounen, genre_Romance, genre_Slice_of_Life, genre_School, genre_Hentai, genre_Supernatural, type_Music, type_ONA, type_OVA, type_Special, type_TV.

The data was split into training and testing sets using an 80:20 stratified split (random_state = 42). Training set: 9,651 records (9,149 class 0; 502 class 1). Test set: 2,413 records (2,288 class 0; 125 class 1).

Gaussian Naive Bayes (GaussianNB) was selected because it is a fast probabilistic classifier for numeric features. It assumes that features are conditionally independent given the class and that each feature follows a Gaussian distribution within each class. In practice, metadata features may be correlated, so the independence assumption is only approximate. Scaling is not required for this model.

Settings tested

The var_smoothing parameter was tested at 1e-11, 1e-9, 1e-8, 1e-7 and 1e-5. It adds numerical stability to the feature variance estimates. The model-selection rule chose the first setting with the highest F1-score; therefore 1e-8 was selected. The settings 1e-8, 1e-7 and 1e-5 tie on F1, while 1e-5 gives the highest AUC.

var_smoothing	Accuracy	Precision	Recall	F1	AUC
1e-11	0.8981	0.2381	0.4400	0.3090	0.7992
1e-09	0.9416	0.4000	0.2560	0.3122	0.8112
1e-08	0.9428	0.4177	0.2640	0.3235	0.8238
1e-07	0.9428	0.4177	0.2640	0.3235	0.8307
1e-05	0.9428	0.4177	0.2640	0.3235	0.8465

Final selected model: GaussianNB(var_smoothing = 1e-8).

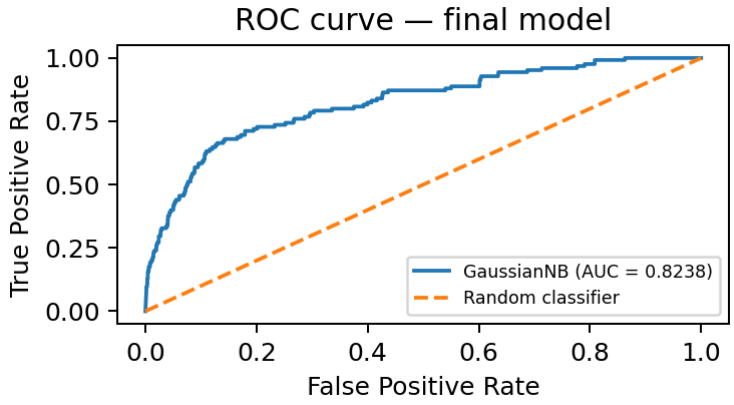
5. Results: Comparison, Confusion Matrix and ROC

Metric	Baseline 1e-9	Final 1e-8
Accuracy	0.9416	0.9428
Precision	0.4000	0.4177
Recall	0.2560	0.2640
F1-score	0.3122	0.3235
Specificity	0.9790	0.9799
ROC-AUC	0.8112	0.8238

Final confusion matrix: [[2242, 46], [92, 33]] (rows = actual class; columns = predicted class).

Actual / Predicted	Not Top	Top Anime
Not Top	2242 — True Negative	46 — False Positive
Top Anime	92 — False Negative	33 — True Positive

The final model correctly identifies 2242 Not Top Anime and 33 Top Anime. It incorrectly labels 46 Not Top Anime as Top Anime and misses 92 actual Top Anime. Specificity is high (97.99%), but recall is low (26.40%), showing the model is much better at recognizing the majority class than finding all Top Anime.



The ROC curve plots the true-positive rate against the false-positive rate across classification thresholds. The final model's ROC-AUC is 0.8238; a value above 0.5 indicates better ranking ability than random guessing.

6. Conclusion and What I Learned

The project built a binary classifier for identifying anime with a rating of at least 8. After cleaning, 12,064 records were available. Feature engineering created genre_count and one-hot genre/type indicators, and SelectKBest reduced the candidate inputs to 15 features. The final GaussianNB model achieved 94.28% accuracy, 0.3235 F1-score and approximately 0.8238 ROC-AUC on the held-out test set.

The high accuracy should be interpreted carefully because only 5.2% of the data belongs to the Top Anime class. The final model's recall of 26.40% means it detects only about one quarter of the actual Top Anime records at the default threshold. A future improvement would be to test class weighting or resampling, adjust the classification threshold, and compare additional algorithms if the assignment permits.

What I learned

• How to inspect a dataset, build a data dictionary and check missing values and duplicates.
• Why target leakage must be prevented by excluding rating from model inputs.
• How to clean mixed numeric/text values such as “Unknown” episodes.
• How feature engineering, one-hot encoding and SelectKBest can turn raw metadata into model-ready inputs.
• Why a stratified train/test split preserves the class ratio.
• Why accuracy is not enough for imbalanced data, and how precision, recall, F1, specificity, confusion matrix and ROC-AUC provide a fuller evaluation.
• How GaussianNB works and why its conditional-independence assumption may not fully hold for real metadata.

Overall, this project helped me understand the complete machine-learning workflow, from data preparation and exploration to model tuning and evaluation. The model is a useful baseline, but improving Top Anime recall is the main next step.

Dataset: anime.csv supplied for this assignment. Implementation and reported results are based on the repaired project notebook. Project ID P45 follows the assignment notebook naming convention (RollNo_P45_Name).